The Sign Premise Is a State Clause: Pumped Baths, the de Sitter Thermostat, and the Limits of *Scale Without Law*

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Abstract

Scale Without Law (Zenodo 10.5281/zenodo.21016309) closed the nonlocal horn of the covariant modified-inertia problem with a sign theorem: a worldline coupled to a **passive** ghost-free bath acquires a mass shift $\delta m = 2\int \rho(\omega)/\omega^2 \, d\omega \geq 0$ — the *anti*-MOND sign. This note sharpens that theorem in both directions, adversarially verified, and reports the results with their edges attached. **First, the positive edge:** the theorem's operative premise is the bath's *state*, not its ghost-freedom. An explicitly ghost-free two-level bath prepared in a population-inverted (non-KMS) steady state flips the sign of the response — $\delta m \propto (p_g - p_e)$ — with total-system unitarity intact; this is laboratory physics (gain-assisted anomalous dispersion), and the theorem's spectral density generalizes in a stationary non-equilibrium steady state to $\rho_{NESS}(\omega) = \Sigma|B_{nm}|^2(p_n - p_m)\delta(\omega - \omega_{mn})$, whose sign at each frequency is a population difference. Ghost-freedom and unitarity therefore do **not** force the anti-MOND sign; passivity does. **Second, a new rigidity wall:** for *linear* coupling to a **free** bosonic field the commutator $[Q(t), Q(0)]$ is a c-number, so the retarded response is *state-independent* — no thermal, squeezed, or pumped Gaussian state of a free field can flip the sign; pumping a free field only adds noise, never gain. The state-clause escape thus requires an **anharmonic** medium, and the de Sitter–Unruh free-field bath — this framework's own premise — cannot carry the MOND sign at *any* occupation. **Third, the thermostat theorem:** the hope that "the de Sitter horizon is the pump" fails four independent ways. The Gibbons–Hawking state is KMS at $T_{dS} = \hbar H_\Lambda/2\pi k_B$ — exactly the passivity clause, with zero extractable work in any cyclic process (Pusz–Woronowicz); its occupation in the galactic dynamical band ($\sim 10^2 - 10^3 \, H_0$) is smaller than 10^{-900}; the real universe's departure from exact de Sitter supplies no in-band inversion (short of the minimum pumping rate by a factor $\sim 10^{10}$ even under maximally generous assumptions); and counterfactually, even a population-inverted mode *at* the horizon's own frequency produces a dispersive mass shift of the **anti**-MOND sign in the band above it. The clean statement: **the very thermality that lets T_{dS} set the *scale* a_0 forbids the horizon from supplying the *sign*.** Fourth, physical gain saturation cannot produce the MOND interpolation $\mu(a/a_0)$: the required saturation variable runs from $\sqrt{g}$ to linear-in-g where physical saturation is quadratic, both asymptotic exponents come out wrong, exact cancellation ($c = 1$) would be an unenforced equivalence-principle fine-tuning, and the saturation scale is a free medium constant — $a_0 = cH_\Lambda/Z$ is not delivered. The net effect is a merger: the sign-theorem escape and the previously-published "galactic pump" specification collapse into **one** precisely-specified, currently unoccupied requirement — a universal, anharmonic, cosmologically-regulated medium with a band-limited time-reversal-odd coupling — whose class-level falsifier is already scheduled: wide-binary dynamics at Gaia DR4, where this class predicts $\gamma \to 1$. *Scale Without Law*'s trichotomy stands unweakened on its own turf; its nonlocal horn is now stated exactly.

1. The theorem, and what its premise actually is

The sign theorem of *Scale Without Law* reads: for a worldline whose inertia is dressed by a linear-response kernel from a ghost-free bath in a passive state, the induced mass shift is $\delta m = 2\int_0^\infty \rho(\omega)/\omega^2\, d\omega$ with $\rho(\omega) \geq 0$ — inertia can only *increase.* Modified-inertia MOND needs the opposite ($m_{eff}/m = \sqrt{g/(g+a_0)} \leq 1$), so the horn closed against passive baths, leaving "an active drive" as the flagged crack.

The present note asks the sharp question: which premise carries the sign — ghost-freedom, or passivity? The answer is unambiguous and verified in two independent conventions (explicit Heisenberg-picture two-level matrices; velocity-coupling equation of motion): **passivity**. For a ghost-free two-level bath degree of freedom B with gap ω_0 and populations (p_g, p_e),

$$\chi_R(0) = 2|b|^2(p_g - p_e)/\omega_0, \quad \delta m = 2|b|^2(p_g - p_e)/\omega_0^2,$$

and a population-inverted steady state ($p_e > p_g$ — definitionally non-KMS, since KMS at any temperature gives $p_g - p_e = \tanh(\beta\omega_0/2) > 0$) **flips the sign** while every mode of the total system remains ghost-free and unitary; the energy is paid by whatever maintains the inversion. This is not exotic: it is the physics of gain media, where anomalous dispersion with negative effective response is measured routinely (Wang, Kuzmich & Dogariu 2000). In a general stationary NESS the theorem's spectral density is replaced by

$$\rho_{NESS}(\omega) = \Sigma_{nm} |B_{nm}|^2 (p_n - p_m)\, \delta(\omega - \omega_{mn}),$$

whose sign at each frequency is the population difference across the connected pairs. **Ghost-freedom never enters the sign.** The banked theorem is thus a *state* theorem — exactly correct on its stated (passive) premises, at every temperature, and silent the moment the state is driven.

2. The rigidity wall: free fields are state-blind

The counterexample above uses a *finite-level* (anharmonic) bath unit, and this is not incidental. For **linear coupling to a free bosonic field**, the field commutator $[Q(t), Q(0)]$ is a c-number — proportional to the identity operator — so the retarded response χ_R, and with it δm, is **independent of the field's state**. No thermal state, no squeezed state, no pumped or expanding Gaussian state of a free field changes the sign; pumping a free field feeds the *symmetric* (noise) correlator only — it heats the worldline without ever supplying gain. (Verified numerically on truncated boson algebras and analytically.)

Two consequences. (i) The state-clause escape of §1 **requires an anharmonic medium** — level structure or self-interaction in the bath sector is not optional. (ii) The de Sitter–Unruh bath of this framework's own founding picture is a free field in a maximally symmetric state: **it cannot carry the MOND sign at any occupation whatsoever.** The state clause is true — and vacuous for the framework's own bath. Whatever medium might realize the escape, it is a *new posit*, not the vacuum already in the theory.

3. The thermostat theorem: the horizon sets the scale and cannot supply the sign

It is tempting — we were tempted — to identify the required drive with the de Sitter horizon itself: a "permanent free-energy source crossing every worldline." Adversarial computation kills this cleanly, four independent ways:

1. **Thermodynamics.** The Gibbons–Hawking state of the static patch is KMS at $T_{dS} = \hbar H_\Lambda/2\pi k_B \approx 2.2\times10^{-30}$ K (Gibbons & Hawking 1977; Bros–Epstein–Moschella). KMS *is* the passivity premise of the theorem, and a single KMS bath yields **zero extractable work** in any cyclic process (Pusz & Woronowicz 1978). Exact de Sitter is a maximum-entropy equilibrium — a thermostat. Free energy requires a second reservoir or a non-KMS state; exact de Sitter supplies neither.
2. **Spectral emptiness.** Galactic orbital dynamics lives at $\omega \sim 4\times10^2 - 2\times10^3\ H_0$. The horizon's thermal occupation there is $n = (e^{2\pi\omega/H_\Lambda} - 1)^{-1}$, i.e. $\log_{10} n \approx -1200$ to -2700. The bath is empty in-band by more than nine hundred orders of magnitude; there is no power to deliver, inverted or otherwise.
3. **No in-band non-KMS channel.** The real universe departs from de Sitter ($|\dot{H}|/H_0^2 \approx 0.5$), but adiabatically at band frequencies ($|\dot{H}|/\omega^2 \sim 10^{-7}$): the induced occupations are $\lesssim 10^{-13}$ and *fall* with frequency — passive ordering, no inversion anywhere. Even granting full inversion and order-unity couplings to every such quantum, the available pumping rate falls short of the minimum phase-pinning rate ($\Gamma \geq 3H_0$, from the companion pump-hunt analysis) by a factor $\sim3\times10^{10}$.
4. **The counterfactual.** Grant the impossible — an inverted mode at the horizon's own frequency $\omega_L \sim H_0$, the only place de Sitter has spectral weight. Its dispersive contribution in the galactic band is $\delta m(\omega \gg \omega_L) = A/(\omega_L^2 - \omega^2) > 0$ for gain ($A < 0$): **anti-MOND**. Below-band gain pushes in-band inertia *up*. The horizon cannot pump the MOND sign into galaxies even in principle.

The resolution is symmetrical and, we think, the honest heart of this note: **the Gibbons–Hawking thermality is precisely why $T_{dS} \sim a_0/c$ works as a *scale* — and precisely why the horizon cannot work as a *pump*.** Scale yes; sign no. The framework keeps the first and must look elsewhere — outside its own vacuum — for the second.

4. Saturation does not shape μ

Deep-MOND stability sits at a cliff: $m_{eff} \to 0$ approaches the runaway threshold $|\delta m| = m$, so the physical realization of any pumped kernel is carried entirely by gain *saturation*. Could saturation itself supply the MOND interpolation? No. Against the physical saturation laws (homogeneous and inhomogeneous two-level broadening): the deep-MOND exponent comes out 2 where μ requires ½; the Newtonian tail comes out g^{-2} where μ requires g^{-1} (the inhomogeneous case can match the tail only, at one tuned condition); reaching $m_{eff} \to 0$ at all requires the cancellation $c \equiv \lambda^2|\chi_B(0)|/m = 1$ *identically for every body* — an unenforced equivalence-principle fine-tuning; and the saturation scale is set by pump rate, linewidth and dipole — free medium constants with no route to $a_0 = cH_\Lambda/Z$. The required saturation variable, solved exactly, runs as $\sqrt{g}$ at small g and linearly at large g, where physical saturation is quadratic in amplitude. **$\mu(a/a_0)$ must be inserted**

by hand; the coefficient is not delivered. This repeats, in the driven setting, the lesson of *Scale Without Law*'s decoy-family test.

5. The merged specification, and its falsifier

These results collapse two previously separate open items — the sign-theorem's "active drive" crack and the published galactic-pump specification — into **one**: the only remaining route to a covariant modified-inertia MOND through this horn is a *universal, anharmonic, cosmologically-regulated medium*, maintained in an inverted steady state, with a band-limited, time-reversal-odd coupling to baryon velocity, a saturation law shaped (by hand) to μ, an exact c = 1 equivalence-principle condition, and an externally supplied scale a_0. Every named candidate for such a medium has been killed in the companion hunt; the specification is precise and **unoccupied**. It is not a theorem of impossibility — it is a bill of requirements, each item computable. The class as a whole carries a scheduled falsifier it cannot evade: a worldline-band-limited kernel leaves Solar-System dynamics Newtonian (γ = 1 exactly — the Cassini gate is passed trivially) and therefore predicts **no MOND excess in wide binaries: γ → 1 at Gaia DR4.** A confirmed wide-binary boost at the modified-gravity level would kill this entire class; a Newtonian null would kill the framework's galaxy-scale phenomenology instead. Either way, the data decide within the decade.

6. What stands

Scale Without Law's trichotomy is not weakened on its own turf — the passive theorem is confirmed exactly, at every temperature, in two conventions. What this note adds is its exact boundary: the sign premise is the bath's state; the escape through that boundary requires an anharmonic medium free fields cannot imitate; the framework's own vacuum is disqualified from supplying it by the same thermality that makes its scale work; and the surviving route is a single, sharp, presently empty specification with a Gaia-DR4 falsifier attached. We claim no pump, no derived μ, no derived a_0, and nothing beyond gravity. The value of the note is that the last open horn of the modified-inertia problem is now stated with edges instead of hope.

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